BDA Week 12 — Advanced Link Analysis

Topic-Sensitive PageRank · Web Spam · TrustRank · Spam Mass · HITS

BDA Final Exam Prep · Comprehensive Tutor

Week 12 · Advanced Link Analysis

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> Builds directly on W8-9 PageRank. Make sure you can recite $r=\beta Mr+(1-\beta)/N\cdot\mathbf{1}_N$ from memory before starting this document.

§1 — Beginning Key Notes (Study Compass)

The 9 ideas you must walk into the exam owning:

1. **Three weaknesses of vanilla PageRank.** (1) generic popularity misses topic-specific authorities; (2) single notion of importance — HITS provides hubs vs. authorities; (3) gameable through link spam — TrustRank fixes this.
2. **Topic-Sensitive PageRank.** Change the teleport target. Instead of jumping uniformly to ANY page, jump uniformly to a small *topic set* S. One PageRank vector per topic.
3. **Topic-PR formula.** $A_{ij}=\beta M_{ij}+(1-\beta)/|S|$ if $j\in S$, else βM_{ij}. Equivalently: $r=\beta Mr+(1-\beta)v_S$, where v_S has $1/|S|$ in positions S and 0 elsewhere.
4. **Term spam.** Stuffing pages with hidden keywords. Google's defense: trust anchor-text from incoming links, not page's own words.
5. **Link spam & spam farms.** Three page types — *inaccessible* / *accessible* / *owned*. Spammer creates M owned pages all linking to target t and sneaks a tiny external boost x from accessible pages.
6. **Spam farm formula.** $y=x/(1-\beta^2)+c\cdot M/N$, where $c=\beta/(1+\beta)$. For $\beta=0.85$, the multiplier $1/(1-\beta^2)\approx 3.61$.
7. **TrustRank.** Topic-sensitive PageRank with the teleport set = trusted human-vetted seed pages (typically `.edu`, `.gov`, `.mil` domains).
8. **Spam Mass.** $SM(p)=(r_p-r^+_p)/r_p$ where r^+_p is TrustRank. Pages with mass close to 1 are spam.
9. **HITS.** Two scores — h_i (hub) and a_j (authority). Iterate $h\leftarrow Aa$ and $a\leftarrow A^T h$ with normalization. Different from PageRank; both are exam-bait concepts.

§2 — Three Limitations of Generic PageRank

Vanilla PageRank gives a single rank per page reflecting overall web-graph importance. This works well for queries like "Stanford" but fails for three reasons:

(1) No topic awareness. A query for "Trojan" maps to three different topics — sports (USC Trojans), history (Trojan war), or computer security (Trojan malware). Generic PageRank cannot prefer the right interpretation given the user's actual interest.

(2) Single notion of importance. Some pages are *good answer pages* (authoritative content), and some are *good index pages* (useful pointers). PageRank conflates these. The HITS algorithm splits importance into two scores: **hubs** (pages that point to many authorities) and **authorities** (pages pointed to by many hubs).

(3) Spam vulnerability. The algorithm is purely structural — it cannot tell whether an incoming link is genuine or spam-farmed. The fix is TrustRank.

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§3 — Topic-Sensitive PageRank

Idea. Recall the random surfer with teleport probability $1-\beta$. Vanilla PageRank teleports to a uniformly random page (every page equally likely). **Topic-sensitive PageRank** instead teleports to a uniformly random page from a *topic-relevant set* S — pages known to be about a particular topic, for example all DMOZ-classified "sports" pages.

Matrix formulation

Define the topic-sensitive Google matrix:

$$A_{ij} = \beta M_{ij} + \frac{1-\beta}{|S|} \;\;\text{if } j \in S, \quad A_{ij} = \beta M_{ij} \;\;\text{otherwise}$$

Equivalently — and this is the form most often used on exams:

$$r = \beta M r + (1 - \beta) \cdot v_S$$

where v_S is a vector with $1/|S|$ in positions corresponding to pages in S, and 0 elsewhere. Plugging $|S| = N$ and $v_S = (1/N)\mathbf{1}_N$ recovers vanilla PageRank — Topic-PR is a strict generalization.

One vector per topic. Compute and store $r_{S_1}, r_{S_2}, \ldots$ for the 16 DMOZ top-level categories (arts, business, sports, …). At query time, pick the right vector based on inferred topic.

> **Algorithm change — only one line.** The complete algorithm from W8-9 changes by exactly one line: replace the *uniform redistribution* step with: $r^{new}_j = r'^{new}_j + (1-S)/|S|$ if $j \in S$, else r'^{new}_j. Everything else stays identical.

Slide example (4-node graph)

The slides show a 4-node graph with edges $1 \to 2$, $1 \to 3$, $2 \to 3$, $3 \to 1$, $3 \to 4$, $4 \to 1$, $4 \to 2$. Out-degrees: $d_1 = 2$, $d_2 = 1$, $d_3 = 2$, $d_4 = 2$. Selected slide answers: - $S = \{1\}$, $\beta = 0.8$: converged $r \approx (0.29, 0.11, 0.32, 0.26)$ - $S = \{1, 2, 3, 4\}$, $\beta = 0.8$: $r \approx (0.13, 0.10, 0.39, 0.36)$ — equivalent to vanilla PageRank - $S = \{1\}$, $\beta = 0.7$: $r \approx (0.39, 0.14, 0.27, 0.19)$ — lower β concentrates more rank in the seed.

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§4 — Web Spam: Term Spam & Google's Defense

Spamming is any deliberate action to boost a page's search-engine ranking beyond its real value. **Spam** is the resulting page. Approximately 10–15% of web pages are spam.

First-generation spam (term spam)

Early search engines ranked by word-frequency on the page. Spammers exploited this in two ways:

- **Keyword stuffing.** Add the target keyword (e.g. "movie") thousands of times, with text color matching the background so users don't see it but the crawler does.
- **Top-result mimicry.** Run the query "movie" yourself, see which page comes first, copy its content invisibly into your own page.

Google's term-spam fix

Believe what others say about you, not what you say about yourself. Specifically, weight the **anchor text** of incoming links — the visible text inside `<a>` tags pointing TO your page — much more than your page's own body text. Combined with PageRank, this surfaced authentic, well-cited pages while burying term-spam farms.

> **Why anchor-text works.** If 1000 pages link to a movie review with anchor text "great movie", that's 1000 independent judgments. The reviewed page can't game this without compromising the linking pages — which themselves have to convince their own users to keep visiting. Distributed truth-telling.

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§5 — Link Spam & Spam Farms

Once Google adopted PageRank + anchor text as the ranking signal, spammers shifted to **link spam**: artificially constructing link structures to boost the PageRank of a target page t.

Three page types from the spammer's view

- **Inaccessible** pages — controlled by others, no influence possible.
- **Accessible** pages — e.g. blog comment sections where the spammer can post a link.
- **Owned** pages — completely under the spammer's control, possibly across many domains.

Spam-farm analysis

Let: - x = total PageRank contributed to t by accessible pages (small, but nonzero). - y = PageRank of target t (what we want to estimate). - M = number of farm pages owned by spammer. - N = total pages on the web.

Each farm page receives $\beta y/M$ from t (which splits its rank among M owned out-links) plus the teleport $(1-\beta)/N$. So:

$$\mathrm{rank}_{\mathrm{farm}} = \beta \cdot \frac{y}{M} + \frac{1-\beta}{N}$$

Then t's rank is x (from accessible) + $\beta M \cdot \mathrm{rank}_{\mathrm{farm}}$ (from M farm pages) + $(1-\beta)/N$ (teleport):

$$y = x + \beta M \cdot \left(\frac{\beta y}{M} + \frac{1-\beta}{N} \right) = x + \beta^2 y + \beta(1-\beta) \frac{M}{N}$$

Solving for y (and dropping the $(1-\beta)/N$ as negligible):

$$y = \frac{x}{1 - \beta^2} + c \cdot \frac{M}{N}, \quad c = \frac{\beta}{1+\beta}$$

Multiplier. For $\beta = 0.85$, $1/(1-\beta^2) \approx 3.61$. Even a small accessible-PR contribution $x = 0.01$ becomes $y \approx 0.036 + c \cdot M/N$. By making M huge, the spammer can push y arbitrarily high — that is the spam farm's "multiplier effect".

> **EXAM TRAP.** The denominator is $1-\beta^2$, not $1-\beta$. The β^2 comes from the round-trip rank flow: $t \to \mathrm{farm} \to t$.

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§6 — TrustRank: Combating Link Spam

Core insight (approximate isolation). Trustworthy pages tend not to link to spam pages. If we can identify a small set of **trusted seed pages** by hand, we can propagate their trust outward through the link graph — pages reachable from trusted seeds in few hops are likely good; pages far from any trusted seed are likely spam.

Algorithm

TrustRank is exactly Topic-Sensitive PageRank with the topic set S = trusted seed pages. Each page receives a trust value $t_p \in [0, 1]$.

- Set $t_p = 1$ for every trusted seed page, 0 elsewhere initially.
- Iterate: each page p with out-link set O_p contributes $\beta t_p / |O_p|$ to each $q \in O_p$.
- Sum incoming contributions per page; converge as in PageRank.

Two effects you should be able to articulate on the exam

- **Trust attenuation** — trust decreases with graph distance from seeds, because at every hop only a β fraction is propagated and the rest leaks into teleport.
- **Trust splitting** — a page with many out-links splits its trust thinly across them. Captures the intuition that an editor with 1000 links per page reviews each less carefully.

Picking the seed set

Two competing goals — small enough that humans can vet every page, yet broad enough that every legitimate web region is reachable in few hops. Two practical tactics: - Take the top-k pages by vanilla PageRank (theory: spam can't reach the very top). - Restrict to controlled-membership domains: `.edu`, `.gov`, `.mil`.

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§7 — Spam Mass: Estimating Fractional Spam Contribution

Complementary view. TrustRank tells us how much of a page's rank comes from *trusted* sources. Spam Mass asks the converse: how much of a page's rank comes from *spam*?

Let: - r_p = vanilla PageRank of page p. - r^+_p = TrustRank of p (= topic-sensitive PR with teleport into trusted seed set). - $r^-_p = r_p - r^+_p$ = rank contributed by non-trusted (likely-spam) sources.

Spam Mass of page p:

$$\mathrm{SpamMass}(p) = \frac{r^-_p}{r_p} = \frac{r_p - r^+_p}{r_p}$$

Spam Mass close to 1 means almost all of p's rank comes from non-trusted sources → p is likely spam. Spam Mass close to 0 means most of p's rank comes from trusted sources → p is legitimate. Pick a threshold (e.g. 0.5) and label everything above it as spam.

> **Why use both r and r⁺?** If we only had TrustRank, we'd label every page far from the seed set as spam — including perfectly legitimate isolated communities. Spam Mass instead asks: *relative to its overall rank, how much of this page's rank is attributable to spam-like sources?* Removes the "distance from seed" bias.

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§8 — HITS: Hubs & Authorities (Brief)

HITS (Hyperlink-Induced Topic Search) is the second classical link-analysis algorithm. The slides only mention HITS in passing as "Solution: Hubs-and-Authorities" — but the concept is fair game on conceptual exam questions because it's in MMDS §5.5.

Two scores per page

- **Hub score** h_i — high if i links TO many high-authority pages.
- **Authority score** a_j — high if j is linked FROM many high-hub pages.

Iteration

Let A be the link adjacency matrix ($A_{ij} = 1$ if $i \to j$). Then:

$$h = A \cdot a, \quad a = A^T \cdot h$$

Substituting one into the other: $a = A^T A \cdot a$, $h = A A^T \cdot h$ — so a is the principal eigenvector of $A^T A$ and h is the principal eigenvector of $A A^T$. Power-iterate, normalizing each step (typically L1 or L2).

PageRank vs HITS — common compare-and-contrast question

- **PageRank**: one score per page, computed offline (query-independent), uses teleport.
- **HITS**: two scores per page, traditionally computed on a query-specific subgraph (root + base set), no teleport.
- PageRank uses column-stochastic M; HITS uses raw adjacency A.

- PageRank converges from any positive start due to teleport; HITS needs explicit normalization.
- Both are eigenvector-based; both reduce to power iteration.

§9 — Six Worked Numerical Examples

Worked Example 1 — Topic-Sensitive PageRank, one iteration

Problem. 4-node graph (edges 1→21 \to 2, 1→31 \to 3, 2→32 \to 3, 3→13 \to 1, 3→43 \to 4, 4→14 \to 1, 4→24 \to 2). S={1}S = \{1\}, β=0.8\beta = 0.8. Compute one iteration from uniform r(0)=(1/4)**1**r^{(0)} = (1/4)\mathbf{1}.

M (column-stochastic):

	fr 1	fr 2	fr 3	fr 4
row 1	0	0	1/2	1/2
row 2	1/2	0	0	1/2
row 3	1/2	1	0	0
row 4	0	0	1/2	0

Mr(0)M r^{(0)}**:** - r1′=0+0+(1/2)(1/4)+(1/2)(1/4)=0.250r'_1 = 0 + 0 + (1/2)(1/4) + (1/2)(1/4) = 0.250 - r2′=(1/2)(1/4)+0+0+(1/2)(1/4)=0.250r'_2 = (1/2)(1/4) + 0 + 0 + (1/2)(1/4) = 0.250 - r3′=(1/2)(1/4)+1·(1/4)+0+0=0.375r'_3 = (1/2)(1/4) + 1 \cdot (1/4) + 0 + 0 = 0.375 - r4′=0+0+(1/2)(1/4)+0=0.125r'_4 = 0 + 0 + (1/2)(1/4) + 0 = 0.125

βMr(0)\beta M r^{(0)}**:** 0.8·(…)=(0.200,0.200,0.300,0.100)0.8 \cdot (\ldots) = (0.200, 0.200, 0.300, 0.100). Sum S=0.800S = 0.800.

Add teleport into S={1}S = \{1\}**:** (1−β)/|S|=0.2/1=0.2(1-\beta)/|S| = 0.2/1 = 0.2 added to node 1 only.

r(1)=(0.400,0.200,0.300,0.100)r^{(1)} = (0.400, 0.200, 0.300, 0.100). Sum =1.000= 1.000 ✓.

Node 1 jumps from 0.25 to 0.40 — teleport mass concentrated there.

Worked Example 2 — Topic set with two pages

Problem. Same 4-node graph, S={1,2}S = \{1, 2\}, β=0.8\beta = 0.8.

βMr(0)=(0.200,0.200,0.300,0.100)\beta M r^{(0)} = (0.200, 0.200, 0.300, 0.100) (same as WE 1).

Teleport: (1−β)/|S|=0.1(1-\beta)/|S| = 0.1 to nodes 1 AND 2.

r(1)=(0.300,0.300,0.300,0.100)r^{(1)} = (0.300, 0.300, 0.300, 0.100). Sum =1.000= 1.000 ✓.

Worked Example 3 — Spam-farm multiplier in numbers

Problem. Spammer creates M=105M = 10^5 owned pages, all linking back to target tt. External PR contribution x=10−4x = 10^{-4}. N=109N = 10^9. β=0.85\beta = 0.85.

1−β2=0.27751 - \beta^2 = 0.2775. So x/(1−β2)=10−4/0.2775≈3.604×10−4x/(1-\beta^2) = 10^{-4}/0.2775 \approx 3.604 \times 10^{-4}.

c=0.85/1.85≈0.459c = 0.85/1.85 \approx 0.459. c·M/N=0.459·105/109=4.59×10−5c \cdot M/N = 0.459 \cdot 10^5/10^9 = 4.59 \times 10^{-5}.

y≈3.604×10−4+4.59×10−5≈4.06×10−4y \approx 3.604 \times 10^{-4} + 4.59 \times 10^{-5} \approx 4.06 \times 10^{-4}.

Sanity check. Without the farm, tt's rank would be ≈x=10−4\approx x = 10^{-4}. The farm amplifies it by ∼4×\sim 4 \times. With M=106M = 10^6, c·M/Nc \cdot M/N grows to 4.59×10−44.59 \times 10^{-4} — farm structure dominates.

Worked Example 4 — TrustRank, one iteration

Problem. 5 pages: A,B,C,D,EA, B, C, D, E. Edges: A→BA \to B, B→AB \to A, B→CB \to C, C→DC \to D, D→ED \to E, E→DE \to D. Trusted seed set {A}\{A\}. β=0.85\beta = 0.85.

Out-degrees: dA=1d_A = 1, dB=2d_B = 2, dC=1d_C = 1, dD=1d_D = 1, dE=1d_E = 1. r(0)=(0.2)**1**r^{(0)} = (0.2)\mathbf{1}.

Mr(0)M r^{(0)}**:** - rA′=(1/2)(0.2)=0.100r'_A = (1/2)(0.2) = 0.100 (only B→AB \to A) - rB′=1·0.2=0.200r'_B = 1 \cdot 0.2 = 0.200 - rC′=(1/2)(0.2)=0.100r'_C = (1/2)(0.2) = 0.100 - rD′=1·0.2+1·0.2=0.400r'_D = 1 \cdot 0.2 + 1 \cdot 0.2 = 0.400 (C→DC \to D and E→DE \to D) - rE′=1·0.2=0.200r'_E = 1 \cdot 0.2 = 0.200

βMr(0)\beta M r^{(0)}**:** 0.85·(…)=(0.085,0.170,0.085,0.340,0.170)0.85 \cdot (\ldots) = (0.085, 0.170, 0.085, 0.340, 0.170). S=0.850S = 0.850.

Teleport into {A}\{A\}**:** (1−β)/|S|=0.15/1=0.15(1-\beta)/|S| = 0.15/1 = 0.15 to A only.

r(1)=(0.235,0.170,0.085,0.340,0.170)r^{(1)} = (0.235, 0.170, 0.085, 0.340, 0.170). Sum =1.000= 1.000 ✓.

A's rank jumps because it's the only teleport target.

Worked Example 5 — Spam Mass calculation

Problem. rp=0.012r_p = 0.012, rp+=0.003r^+_p = 0.003. (a) Compute Spam Mass. (b) Threshold 0.5: spam?

1. rp−=0.012−0.003=0.009r^-_p = 0.012 - 0.003 = 0.009. SpamMass =0.009/0.012=0.75= 0.009/0.012 = 0.75.
2. 0.75>0.50.75 > 0.5 → **YES, flagged as spam**. ¾ of its rank is from non-trusted sources.

Worked Example 6 — HITS, two iterations

Problem. 3 pages with adjacency: 1→21 \to 2, 1→31 \to 3, 2→32 \to 3, 3→13 \to 1. Compute h,ah, a for two iterations from h(0)=a(0)=(1,1,1)h^{(0)} = a^{(0)} = (1, 1, 1). Normalize by L1 each step.

A=(011001100)A = \begin{pmatrix} 0 & 1 & 1 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} (rows = source). AT=(001100110)A^T = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix}.

Iter 1. a=ATh=(h3,h1,h1+h2)=(1,1,2)a = A^T h = (h_3, h_1, h_1 + h_2) = (1, 1, 2). L1 = 4. a=(0.25,0.25,0.50)a = (0.25, 0.25, 0.50).

h=Aa=(a2+a3,a3,a1)=(0.75,0.50,0.25)h = A a = (a_2 + a_3, a_3, a_1) = (0.75, 0.50, 0.25). L1 = 1.5. h=(0.500,0.333,0.167)h = (0.500, 0.333, 0.167).

Iter 2. a=ATh=(h3,h1,h1+h2)=(0.167,0.500,0.833)a = A^T h = (h_3, h_1, h_1+h_2) = (0.167, 0.500, 0.833). L1 = 1.5. a=(0.111,0.333,0.556)a = (0.111, 0.333, 0.556).

h=Aa=(0.889,0.556,0.111)h = A a = (0.889, 0.556, 0.111). L1 = 1.556. h=(0.571,0.357,0.071)h = (0.571, 0.357, 0.071).

Strongest authority: page 3 (linked from 1 and 2). **Strongest hub: page 1** (links to 2 and 3).

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§10 — Practice Questions (15)

Q1 [Numerical · 6 marks]. 4-node graph from WE 1. Compute Topic-PR with S={3}S = \{3\}, β=0.85\beta = 0.85, one iteration from uniform.

Q2 [Numerical · 5 marks]. Same 4-node graph, S={1,2,3,4}S = \{1, 2, 3, 4\} (vanilla equivalent), β=0.8\beta = 0.8. Compute one iteration. Verify the teleport contribution equals (1−β)/N(1-\beta)/N per node.

Q3 [Numerical · 6 marks]. Spammer creates M=5×104M = 5 \times 10^4 pages, x=5×10−5x = 5 \times 10^{-5}, N=109N = 10^9, β=0.85\beta = 0.85. (a) Compute multiplier. (b) Compute yy. (c) What MM to make y≥10−3y \geq 10^{-3}?

Q4 [Numerical · 7 marks]. Five-page graph: A→BA \to B, A→CA \to C, B→CB \to C, B→DB \to D, C→AC \to A, D→ED \to E, E→CE \to C. Trusted seeds {A,C}\{A, C\}, β=0.8\beta = 0.8. One iteration of TrustRank from uniform.

Q5 [Numerical · 4 marks]. rp=0.025r_p = 0.025, rp+=0.020r^+_p = 0.020. (a) SpamMass. (b) Threshold 0.4: flagged? (c) New rp+=0.022r^+_p = 0.022 — SpamMass change?

Q6 [Numerical · 6 marks]. 3-page graph: 1→21 \to 2, 1→31 \to 3, 2→12 \to 1, 2→32 \to 3, 3→23 \to 2. HITS for two iterations, L1-normalize each step. Strongest authority? Strongest hub?

Q7 [Concept · 4 marks]. Topic-PR is a one-line modification of vanilla PR. Give an example where they return strictly different rankings.

Q8 [Concept · 4 marks]. Define trust attenuation and trust splitting. Identify each in the algebra of TrustRank.

Q9 [Concept · 4 marks]. Walk through the spam-farm derivation. Why β2\beta^2? Why drop (1−β)/N(1-\beta)/N? Multiplier for β=0.9\beta = 0.9?

Q10 [Concept · 4 marks]. Compare PageRank, TrustRank, HITS, Spam Mass on (a) per-page output, (b) teleport/normalization, (c) question answered.

Q11 [Concept · 3 marks]. Why is Spam Mass preferable to thresholding TrustRank directly? Concrete misclassification scenario.

Q12 [MCQ · 2 marks]. Topic-PR teleport contribution to j∈Sj \in S each iteration: (A) (1−β)/N(1-\beta)/N (B) (1−β)/|S|(1-\beta)/|S| (C) β/|S|\beta/|S| (D) β(1−β)/N\beta(1-\beta)/N

Q13 [MCQ · 2 marks]. Spam-farm multiplier for β=0.5\beta = 0.5: (A) 1.0 (B) 1.33 (C) 1.6 (D) 2.0

Q14 [True/False · 2 marks]. HITS converges from any positive initial vector without normalization. Justify.

Q15 [MCQ · 2 marks]. Most robust TrustRank seed set strategy: (A) random sample of 1000 (B) top-k by PageRank (C) every page on the web (D) `.edu` / `.gov` / `.mil` only

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§11 — Full Worked Answers

A1. Mr(0)=(0.250,0.250,0.375,0.125)M r^{(0)} = (0.250, 0.250, 0.375, 0.125). βMr=0.85·(…)=(0.2125,0.2125,0.31875,0.10625)\beta M r = 0.85 \cdot (\ldots) = (0.2125, 0.2125, 0.31875, 0.10625). (1−β)/|S|=0.15(1-\beta)/|S| = 0.15 to node 3. r(1)=(0.2125,0.2125,0.46875,0.10625)r^{(1)} = (0.2125, 0.2125, 0.46875, 0.10625).

A2. βMr=(0.200,0.200,0.300,0.100)\beta M r = (0.200, 0.200, 0.300, 0.100). Teleport 0.2/4=0.050.2/4 = 0.05 to every node. r(1)=(0.250,0.250,0.350,0.150)r^{(1)} = (0.250, 0.250, 0.350, 0.150). ✓ each node gets exactly (1−β)/N(1-\beta)/N.

A3. (a) 1/(1−0.7225)=3.6041/(1-0.7225) = 3.604. (b) c=0.4595c = 0.4595. y=5×10−5·3.604+0.4595·5×104/109=1.802×10−4+2.298×10−5≈2.03×10−4y = 5 \times 10^{-5} \cdot 3.604 + 0.4595 \cdot 5 \times 10^4/10^9 = 1.802 \times 10^{-4} + 2.298 \times 10^{-5} \approx 2.03 \times 10^{-4}. (c) Need M≥109·(10−3−1.802×10−4)/0.4595≈1.78×106M \geq 10^9 \cdot (10^{-3} - 1.802 \times 10^{-4})/0.4595 \approx 1.78 \times 10^6 pages.

A4. Out-degrees: dA=2,dB=2,dC=1,dD=1,dE=1d_A = 2, d_B = 2, d_C = 1, d_D = 1, d_E = 1. - rA′=1·0.2=0.200r'_A = 1 \cdot 0.2 = 0.200 (C→A) - rB′=(1/2)(0.2)=0.100r'_B = (1/2)(0.2) = 0.100 (A→B) - rC′=(1/2)(0.2)+(1/2)(0.2)+1·0.2=0.500r'_C = (1/2)(0.2) + (1/2)(0.2) + 1 \cdot 0.2 = 0.500 (A,B,E→C) - rD′=(1/2)(0.2)=0.100r'_D = (1/2)(0.2) = 0.100 (B→D) - rE′=1·0.2=0.200r'_E = 1 \cdot 0.2 = 0.200 (D→E)

βMr=(0.160,0.080,0.400,0.080,0.160)\beta M r = (0.160, 0.080, 0.400, 0.080, 0.160). S=0.880S = 0.880. To preserve ∑=1\sum = 1: use (1−S)/|S|=0.06(1-S)/|S| = 0.06 on A and C. r(1)=(0.220,0.080,0.460,0.080,0.160)r^{(1)} = (0.220, 0.080, 0.460, 0.080, 0.160).

A5. (a) rp−=0.005r^-_p = 0.005. SpamMass =0.005/0.025=0.20= 0.005/0.025 = 0.20. (b) 0.20<0.400.20 < 0.40 → NOT flagged. (c) New rp−=0.003r^-_p = 0.003. New SpamMass =0.12= 0.12. Decreased by 0.08.

A6. A=(011101010)A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}. AT=(010101110)A^T = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}. h(0)=a(0)=(1,1,1)h^{(0)} = a^{(0)} = (1, 1, 1).

Iter 1: a=ATh=(1,2,2)a = A^T h = (1, 2, 2). L1 = 5. a=(0.20,0.40,0.40)a = (0.20, 0.40, 0.40). h=Aa=(0.80,0.60,0.40)h = A a = (0.80, 0.60, 0.40). L1 = 1.80. h=(0.444,0.333,0.222)h = (0.444, 0.333, 0.222).

Iter 2: a=ATh=(0.333,0.667,0.778)a = A^T h = (0.333, 0.667, 0.778). L1 = 1.778. a=(0.187,0.375,0.438)a = (0.187, 0.375, 0.438). h=(0.813,0.625,0.375)h = (0.813, 0.625, 0.375). L1 = 1.813. h=(0.448,0.345,0.207)h = (0.448, 0.345, 0.207).

Strongest authority: page 3. Strongest hub: page 1.

A7. Modify ONE line of the W8-9 algorithm: redistribute leaked mass to SS only. Example: 2-cycle 1→21 \to 2, 2→12 \to 1 + isolated page 3. Vanilla PR: r≈(0.4,0.4,0.2)r \approx (0.4, 0.4, 0.2). With S={3}S = \{3\}: r≈(0.15,0.15,0.70)r \approx (0.15, 0.15, 0.70). Different #1.

A8. Trust attenuation — at each hop, only β\beta fraction propagates → after kk hops, only βk\beta^k survives. **Trust splitting** — page with kk out-links splits trust by 1/k1/k. Captured by 1/di1/d_i in MjiM_{ji}.

A9. y=x+βM(βy/M+(1−β)/N)+(1−β)/Ny = x + \beta M (\beta y / M + (1-\beta)/N) + (1-\beta)/N. Round-trip t→farm→tt \to \mathrm{farm} \to t gives β2\beta^2. (1−β)/N(1-\beta)/N is O(1/N)≈10−9O(1/N) \approx 10^{-9} — negligible. For β=0.9\beta = 0.9: 1/(1−0.81)≈5.261/(1-0.81) \approx 5.26.

A10. - **Per-page output.** PR: 1 score; TrustRank: 1 trust score; HITS: 2 scores; SpamMass: 1 ratio. - **Teleport/normalization.** PR: uniform; TrustRank: trusted seeds; HITS: no teleport, L1/L2 norm; SpamMass: derived (no iteration). - **Question.** PR: “important?”; TrustRank: “trusted?”; HITS: “hub or authority?”; SpamMass: “fraction from spam?”

A11. Direct TrustRank thresholding misclassifies legitimate isolated pages (low r+r^+ regardless of legitimacy). E.g., legit blog with rp=10−4r_p = 10^{-4}, rp+=9×10−5r^+_p = 9 \times 10^{-5}. Threshold 0.001 flags as spam. SpamMass =0.10= 0.10 → not flagged. Normalization removes “distance bias”.

A12. (B) (1−β)/|S|(1-\beta)/|S|.

A13. 1/(1−0.25)=1.3331/(1 - 0.25) = 1.333. **(B)**.

A14. False. h,ah, a explode or vanish without normalization since the eigenvalue of ATAA^T A is generally not 1.

A15. (D). Controlled domains have negligible spam. (B) is reasonable but top-PR pages can include compromised popular sites.

§12 — Ending Key Notes (Revision Cards)

Card	Quick-fact
PR's 3 weaknesses	Topic blind / single score / link-spam vulnerable
Topic-PR formula	$r=\beta Mr+(1-\beta)vSr = \beta M r + (1-\beta) v_S$, $vS[j]=1/\|S\|v_S[j] = 1/\|S\|$ if $j\in Sj \in S$
Topic-PR teleport	$(1-\beta)/\|S\|(1-\beta)/\|S\|$ to every $j\in Sj \in S$, 0 to others
Algorithm change	ONE line: redistribute to S only
Term spam	Hidden keyword stuffing, copying top results
Google's term-spam fix	Trust anchor text, not page's own body
3 page types	Inaccessible / Accessible / Owned
Spam farm goal	Maximize PR of target tt via mutual links with MM owned pages
Spam farm formula	$y=x/(1-\beta2)+c\cdot M/Ny = x/(1-\beta^2) + c \cdot M/N$, $c=\beta/(1+\beta)c = \beta/(1+\beta)$
Multiplier ($\beta=0.85\beta = 0.85$)	$1/(1-\beta2)\approx3.61/(1-\beta^2) \approx 3.6$
TrustRank	Topic-PR with seeds = trusted pages
Trust attenuation	Trust decays $\beta\beta$ per hop
Trust splitting	Page with kk out-links splits trust by $1/k1/k$
Seed set choices	Top-kk by PR, controlled domains (`.edu` / `.gov` / `.mil`)
Spam Mass formula	$SM(p)=(rp-rp+)/rp\mathrm{SM}(p) = (r_p - r^+_p)/r_p$
Spam Mass intuition	Fraction of pp's rank from non-trusted sources
HITS scores	hih_i hub, aja_j authority
HITS iteration	$h\leftarrow Aah \leftarrow A a$, $a\leftarrow ATha \leftarrow A^T h$, normalize each step
HITS eigen-equation	aa = principal eigenvector of $ATAA^T A$
PR vs HITS	PR: 1 score, query-indep, teleport. HITS: 2 scores, query-spec, normalize.

§13 — Formula & Algorithm Reference

Concept	Formula	Use
Topic-Sensitive PR	$r = \beta M r + (1-\beta) v_S$	Bias rank to a specific topic.
Topic-PR teleport	$(1-\beta)/\lvert S\rvert$ $(1-\beta)/\lvert S\rvert$ to seeds only	One-line modification of W8-9 algorithm.
Spam farm rank	$y = x/(1-\beta^2) + (\beta/(1+\beta)) \cdot M/N$	Estimating link-farm effectiveness.
TrustRank	Same as Topic-PR with trusted seeds	Demoting link-spam farms automatically.
Trust propagation	$t_q = \beta \sum_{p \to q} t_p / \lvert O_p\rvert + \mathrm{teleport}$	Per-iteration update.
Spam Mass	$\mathrm{SM}(p) = (r_p - r^+_p)/r_p$	Fractional spam contribution.
HITS authority	$a = A^T h$ → principal eigenvector of $A^T A$	Pages linked-to by good hubs.
HITS hub	$h = A a$ → principal eigenvector of $A A^T$	Pages linking-to good authorities.

Connections to other weeks: - **W8-9 PageRank.** Every algorithm here is a parametrization of W8-9 power iteration. - **W16 Community Detection.** Spectral clustering uses the same eigenvector machinery on the graph Laplacian.

End of W12 Advanced Link Analysis exam-prep document.